

ODEQ Air Quality Forecast Dashboard: Forecasts and Evaluation

October 2026

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DEQ Department of Environmental Quality

Why it was built:

- Found a reliable, crash-proof path to pull *historical + real-time forecast data*, entirely on DEQ's server infrastructure, no local machine dependency, and centralized it on OneDrive. The only workable solution within ODEQ's existing tools.

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Why it was built:

Before this project,

- there was No visibility into IDEQ's ML model
- No evaluation platform
- No multi-source integration
- No way to see forecasted concentration/AQI at our monitor sites; only observed/historical readings existed

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Why it was built:

What's new now

Multi-platform integration: IDEQ's ML model, AIRPACT-5, AIRPACT-6, and NWS forecasts together in one dashboard

A working evaluation platform: can now quantitatively prove model skill (just measured it: IDEQ's ML model beats the alternatives, r=0.93 vs 0.72–0.87)

Automated exceedance alerting via Telegram: currently reaches me, with expansion to other assigned staff as a ready-to-flip-on next step

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Why it was built:

Why timing matters

Staff need all air-quality advisories finalized by 10 AM; there's an 8:30 AM smoke call every other/every morning that depends on it.

For winter stagnation events, NWS doesn't issue stagnation advisories at all; DEQ has to identify and act on those independently.

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Why it was built:

Air stagnation advisory benefit

Early warning for winter stagnation events:

The dashboard lets staff see PM2.5 trending toward an exceedance before it happens; not just after or in-between.

IDEQ's ML model is trained per-monitor, it should perform especially well during winter inversion/stagnation events.

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IDEQ Machine Learning Forecast Model (O_3 & $PM_{2.5}$)

- The IDEQ system is a **hybrid statistical / ML forecast model** that integrates:
  - 1. Historical **observations** (Previous 24-hour observed O_3 / $PM_{2.5}$: t−1 to t−24;
    - Persistence
    - Diurnal pattern recognition
    - Carryover (e.g., residual ozone, smoke accumulation)
  - 2. GEOS-CF O_3 / $PM_{2.5}$ (forecast concentrations used as physics-based baseline, bias-corrected by ML)
  - 3. Meteorological inputs: Temperature (T), Pressure (P), Relative Humidity (RH), PBL Height (PBLH), Wind Speed (WS), Wind Direction (WDIR), Deep Stable Layer (DSL) metrics
- Unlike traditional models such as AIRPACT, the IDEQ model includes:
  - **Real-time bias correction**
  - **Observation updates (persistence)**
  - **Time-of-day adjustments**
- This makes it **operationally adaptive** and often more accurate in the short term

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PM25_AQS_ID	Sitename	Latitude	Longitude	AirNow_site_name
410010004	Baker_City	44.78821945	-117.8445969	Baker City - Forest Service
410090004	Sauvie_Island	45.7685318	-122.7721024	Portland - Sauvie Island
410130100	Prineville	44.29978561	-120.8447723	Prineville - Davidson Park
410170004	Sisters	44.292099	-121.5554962	Sisters Forest Service
410170120	Bend	44.06391525	-121.3125534	Bend - Pump Station
410230002	John_Day	44.41799927	-118.9513016	John Day Dayton St
410250003	Burns	43.58924866	-119.0487213	Burns - Washington St.
410290019	Shady_Cove	42.62322617	-122.8101578	Shady Cove - Shady Cove School
410290201	Medford	42.22989273	-122.7876968	Talent - Rapp Lane
410290203	Ashland	42.19409943	-122.7086029	Ashland Fire Dept
410330036	Cave_Junction	42.1617012	-123.6483536	Cave Junction Forest Service
410330114	Grants_Pass	42.43413925	-123.348465	Grants Pass - Parkside School
410350004	Klamath_Falls	42.19029617	-121.731369	Klamath Falls - Peterson School
410370001	Lakeview	42.18921661	-120.3539658	Lakeview - Center & M
410390059	Eugene2	44.0672226	-123.1413498	Eugene - Highway 99
410390060	Eugene	44.02631378	-123.0837402	Eugene - Amazon Park
410391009	Springfield	44.04669571	-123.0177002	Springfield City Hall
410392013	Oakridge	43.74435043	-122.480484	Oakridge - Willamette Activity Center
410399004	Cottage_Grove	43.79949951	-123.0534897	Cottage Grove City Shops
410430009	Albany	44.61568832	-123.0914536	Albany - Calapooia School
410432002	Sweet_Home	44.39580154	-122.7304993	Sweet Home - Fire Department
410470041	Salem	44.94311523	-123.0059357	Salem - State Hospital
410510080	Portland_SEL	45.49663925	-122.6028748	Portland - SE Lafayette
410590121	Pendleton	45.65222931	-118.8230362	Pendleton - McKay Creek
410610123	La Grande	45.32350159	-118.0777969	La Grande Hall & N Sts
410630001	Enterprise	45.42634964	-117.2960663	Enterprise - US Forest Service
410650007	The_Dalles	45.60242081	-121.2033844	The Dalles - Cherry Heights
410670004	Hillsboro	45.5284996	-122.9723969	Hillsboro - Hare Field
410670005	Tualatin	45.39349747	-122.7478943	Portland Near Road
410670111	Beaverton	45.47019196	-122.8164139	Beaverton - Highland Park School
840410450001	Ontario	44.03336334	-116.9697037	Ontario May Roberts Elementary School

O3_AQS_ID	Sitename	Latitude	Longitude	AirNow_site_name
410090004	Sauvie_Island	45.7685318	-122.7721024	Portland - Sauvie Island
410290201	Medford	42.22989273	-122.7876968	Talent - Rapp Lane
410390060	Eugene	44.02631378	-123.0837402	Eugene - Amazon Park
410391007	Eugene_Saginaw	43.83449936	-123.0353012	Saginaw - Delight Valley School
410470041	Salem	44.94311523	-123.0059357	Salem - State Hospital
410510080	Portland_SEL	45.49663925	-122.6028748	Portland - SE Lafayette
410670005	Tualatin	45.39349747	-122.7478943	Portland Near Road

While some important sites (e.g., the downwind monitor at Carus–Spangler) are not yet included in the trained system.

Dashboard currently supports ML-based predictions for all Oregon sites included in the training dataset, but not for every monitoring site we operate. For ozone predictions, adjacent key locations (Vancouver & Boise) are added.

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IDEQ Machine Learning Forecast Model (O_3 & $PM_{2.5}$)

In this proposal, the IDEQ ML model will be used as:

A core input to the forecasting engine
A benchmark for comparison against AIRPACT and other guidance

Content	Schedule
AM run	Daily 7:30 AM Pacific
PM run	Daily 1:30 PM Pacific

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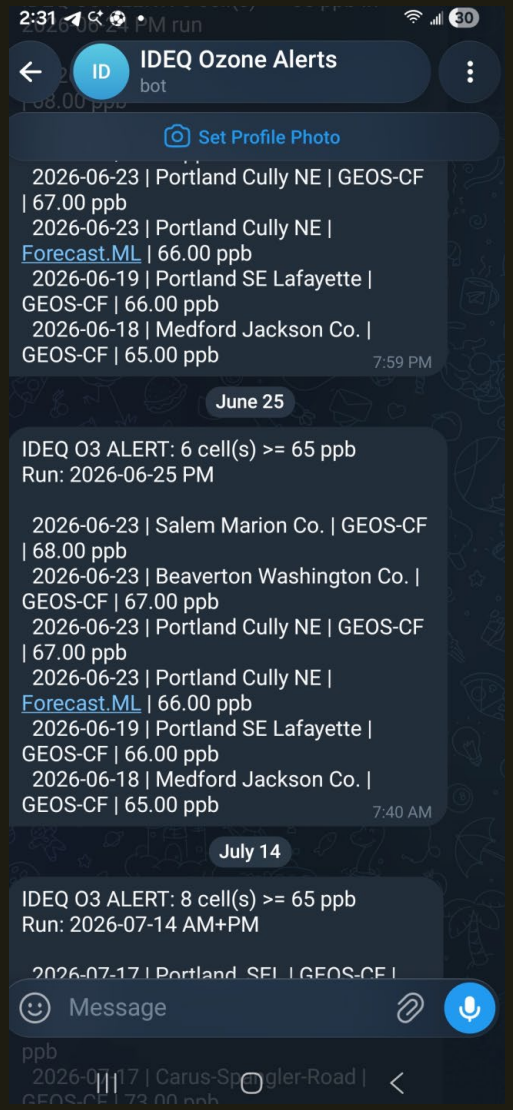
Why it was built:

Exceedance thresholds

Regulatory limit (NAAQS)	Our alert threshold	Why the alert is set below NAAQS	
Ozone	70 ppb (8-hour average)	65 ppb	Gives staff a ~5 ppb early-warning buffer before the forecast actually crosses the legal/health limit
PM2.5	35 µg/m³ (24-hour average)	32 µg/m³	Same idea — a ~3 µg/m³ early-warning margin

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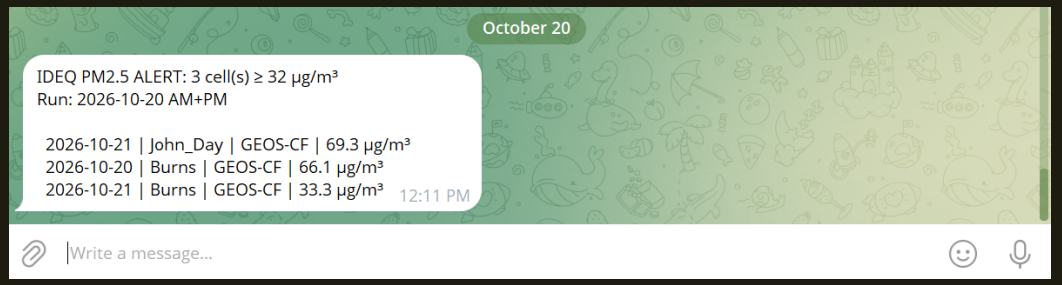
Why it was built:

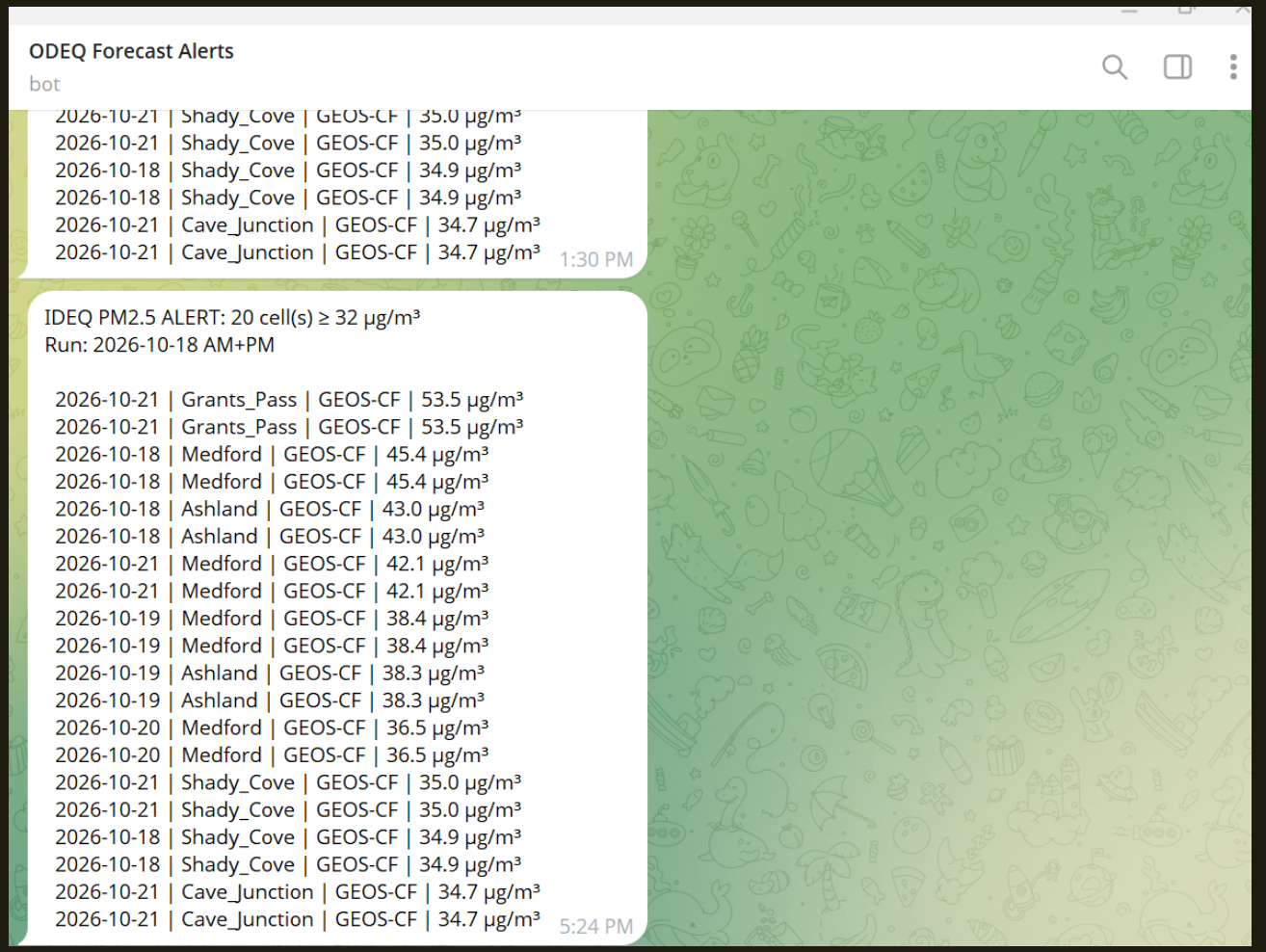




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Why it was built:





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Condition	n days	Mean bias	Median bias	RMSE	Under-predict (N1–N4)	Within ±5 (OK)	Over-predict (P1–P4)
Non-smoke	14	−5.8 ppb	−4.5	7.6	43%	57%	0%
Smoke (WF)	30	−7.2 ppb	−8.0	10.0	60%	33%	7%

DEQ's automated AP6 archive begins mid-October 2026, and the source only serves forward forecasts, so there's no AP6 ozone back-history to evaluate yet.

IDEQ ML underpredicts ozone on high-ozone days, and the underprediction gets worse on wildfire-smoke days.

On smoke days the mean bias deepens from −5.8 to −7.2 ppb:
the share of underprediction days jumps from 43% → 60%;
"good" (±5 ppb) forecasts drop from 57% → 33%;
error (RMSE) rises from 7.6 → 10.0 ppb.

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Next

- AIRPACT-5 data for 2024–2025 is fully available and can be used now.
- AIRPACT-6 is newer, so its historical record is still growing.
- As AP6 continues to accumulate data, it will become possible to assess how well it performs for $PM_{2.5}$ during prescribed burns.

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Next

Comprehensive Evaluation on ozone and pm2.5 in the evaluation (2024-2026 hourly/daily).

Introduce forecast polygons and create forecast map for smoke and evaluation of models in those polygons.

Develop a method that uses satellite-derived VOC and NOx indicators, combined with spatial fire information, Determine whether wildfire emissions shift areas into VOC-limited or NOx-limited regimes